Stranded Capital:
Measuring the Economic Cost of AI Accessibility Exclusion Across the Disability Market

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Abstract

Artificial intelligence platforms are transforming commerce, employment, and civic participation—yet the 1.6 billion people worldwide living with disabilities remain systematically excluded from these gains. We argue this exclusion is not merely an equity concern but a quantifiable macroeconomic failure: spending power that exists but cannot be deployed due to inaccessible AI-mediated interfaces. We introduce the **Stranded Capital Index** (SCI), a novel economic construct measuring the suppression of disability market purchasing power attributable to platform inaccessibility. The SCI is computed as the product of three publicly verifiable variables: WCAG noncompliance rate by sector, estimated disability population disposable income share, and digital commerce penetration. Drawing on the Massenkoff and McCrory [2026] observed-versus-theoretical AI exposure framework, we apply analogous demand-side logic: the gap between theoretical purchasing capacity and observed transaction completion for persons with disabilities (PWDs).

We validate SCI across eight industry sectors using longitudinal WebAIM Million audit data (2019–2026), Bureau of Labor Statistics Current Population Survey disability supplements, and the Accenture Disability Equality Index financial performance dataset. Six analytical conditions (C1–C6) decompose the SCI along dimensions of AI-intensity weighting, regulatory adjustment, firm-level inclusion premia, click-away proxy validation, and temporal drift. Condition C6—the *Temporal Stranding Acceleration* hypothesis—is our novel contribution: the first empirical test of whether AI-driven page complexity growth is outpacing accessibility compliance improvement, producing a net acceleration of stranded capital across time.

Preliminary sector-level analysis finds that e-commerce, financial services, and entertainment show the highest SCI scores, while government shows the largest gap between baseline and regulatory-adjusted SCI—consistent with ADA Title II enforcement effects. The 2026 WebAIM Million reversal (95.9% WCAG failure rate, up from 94.8% in 2025) provides initial support for the C6 temporal acceleration hypothesis. Firms scoring in the top quartile of the Disability Equality Index show revenue multiples of 1.6× (revenue) and 2.6× (net income) versus peers, consistent with a first-mover accessibility premium. These findings have direct implications for AI deployment standards, accessibility-adjusted ESG materiality disclosure, and the equitable distribution of AI productivity gains.

Keywords: stranded capital, disability market, AI accessibility, WCAG compliance, observed exposure, economic exclusion, Stranded Capital Index, disability spending power, first-mover premium, temporal stranding, AI economics, societal impact

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1 Introduction

Every major account of artificial intelligence's economic impact frames the central question as a labor supply problem: which jobs will AI displace, which will it augment, and who will capture the productivity surplus [Massenkoff and McCrory, 2026, Massenkoff et al., 2026]. This framing is incomplete. It attends to the supply side of the economy—what workers can produce with AI—while largely ignoring the demand side: what consumers can purchase through AI-mediated platforms. For the 1.6 billion people worldwide living with disabilities [Return on Disability Group, 2024], the demand-side failure is not marginal. It is structural, measurable, and growing.

The disability market commands between $13 trillion and $23 trillion in annual spending power globally [World Economic Forum, 2025, TriplePundit / Safman, 2025], making it larger than the GDP of China and Japan combined. Yet WebAIM [2026] finds that 95.9% of the top one million website homepages contain detectable WCAG 2 accessibility failures in 2026—a *worsening* from 94.8% in 2025, reversing six consecutive years of marginal improvement. AI-powered platforms are not solving this problem. They are, in many sectors, accelerating it: page complexity has grown 61% over six years as AI-generated content and JavaScript frameworks proliferate, and WCAG failure density is rising in parallel.

The demand-side gap. Massenkoff and McCrory [2026] introduce *observed exposure* as a corrective to purely theoretical AI labor market impact assessments—the gap between what LLMs could theoretically do and what workers actually experience. We apply the same intellectual move to the consumer economy. Disability market spending power is the *theoretical* purchasing capacity. Stranded capital is the gap between that theoretical capacity and *observed* transaction completion. The mechanisms suppressing observed transactions—inaccessible interfaces, absent screen reader support, inaccessible AI-generated ARIA code—are analogous to the automation friction that depresses observed worker augmentation rates. Both represent a failure of AI deployment to realize its theoretical potential.

The microeconomic stakes. Accenture et al. [2023] find that firms leading in disability inclusion generate 1.6× more revenue and 2.6× more net income than industry peers across a 348-company longitudinal panel. BeAccessible [2026] reports that 69% of disabled online consumers abandon transactions at inaccessible sites, representing a UK monthly click-away toll of £2 billion. The United States equivalent, extrapolated from $490 billion in PWD disposable income [TriplePundit / Safman, 2025], suggests a nine-figure monthly economic suppression figure with no existing policy-grade measurement instrument. This paper introduces one.

The regulatory inflection point. The U.S. Department of Justice finalized WCAG 2.1 AA accessibility requirements for state and local government platforms in 2024 [Center for People with Disabilities, 2025]. The European Accessibility Act took effect in June 2025, with EU AI Act high-risk system provisions applying from August 2026 [Clifford Chance, 2024]. These regulatory timelines create an urgent need for a quantitative index that can measure stranded capital before and after compliance interventions, track sectoral variation, and support materiality disclosure under GRI 203 and CSRD ESRS S1. The SCI proposed here is designed to serve all three functions.

We contribute:

1. The first formal definition and computation of **Stranded Capital**—spending power held by PWDs that cannot be deployed due to AI platform inaccessibility—as a measurable economic construct distinct from prior descriptive disability market sizing literature.

2. The **Stranded Capital Index** (SCI): a three-variable index (WCAG non-compliance rate × PWD disposable income share × digital commerce penetration) computed across eight industry sectors under six analytical conditions.

3. A novel **Temporal Stranding Acceleration** condition (C6)—the first empirical test of whether AI-driven page complexity growth is outpacing accessibility compliance gains over the 2019–2026 WebAIM longitudinal series, generating a net increase in stranded capital across time.

4. A publicly released sector-level SCI dataset, scoring rubric, and computation scripts designed to support replication, regulatory benchmarking, and ESG materiality disclosure integration.

This paper proceeds as follows. Section 2 situates our work within the AI economics, disability market, and accessibility compliance literatures. Section 3 presents the SCI formula, data sources, six analytical conditions, failure rubric, and statistical analysis plan. Section 4 documents the anticipated reporting structure for the full analysis. Section 5 develops the policy and design implications. Sections 6–7 address limitations and structural conclusions.

2 Related Work

AI economics and observed vs. theoretical exposure. The foundational empirical framework for this paper is the observed-versus-theoretical exposure distinction introduced by Massenkoff and McCrory [2026]. Their measure combines LLM capability assessments with actual Claude usage data from the Anthropic Economic Index to identify where theoretical AI augmentation potential diverges from what workers actually experience. We extend this logic to the consumer demand side: theoretical disability market purchasing capacity versus observed transaction completion. Massenkoff et al. [2026] further document a skill-biased AI adoption curve in which early, high-skill adopters realize substantially larger augmentation gains than later, less technical adopters—a dynamic that, we argue, applies symmetrically to accessible versus inaccessible consumer platforms. Our paper operationalizes the demand-side analog of observed exposure for the first time.

Disability market economics. The disability market has been characterized descriptively in market sizing reports but not formally modeled as a suppressed economic system. American Institutes for Research [2018] establish baseline estimates of PWD disposable and discretionary income stratified by disability type using American Community Survey microdata, finding $490 billion in US PWD disposable income. Return on Disability Group [2024] scale this globally to $18 trillion in spending power and $2.6 trillion in North American and European disposable income specifically. Accenture et al. [2023] provide the most rigorous financial performance evidence: a 346-company longitudinal panel (2015–2022) demonstrating that DEI-leading firms achieve 1.6× revenue and 2.6× net income relative to non-leaders. None of these studies models the suppression mechanism that creates the gap between theoretical and observed spending. This paper fills that gap.

Web accessibility compliance evidence. WebAIM [2026] provide the most current and comprehensive empirical baseline for platform inaccessibility: 95.9% of the top one million homepages contain detectable WCAG 2 failures in February 2026, with an average of 56.1 errors per page—a 10.1% increase from the 2025 figure of 51 errors per page, reversing six years of marginal improvement. The six categories accounting for 96% of all detected errors—low contrast text, missing alt text, missing form labels, empty links, empty buttons,

missing document language—have remained unchanged for seven consecutive years. This persistence under increasing regulatory pressure is precisely the temporal pattern our C6 condition is designed to measure. WebAIM [2025] establish the 2025 baseline (94.8% failure rate) that enables the 2025–2026 acceleration calculation.

AI accessibility governance. The regulatory landscape for AI accessibility is evolving rapidly and with material ESG implications. Clifford Chance [2024] provide a comprehensive global survey of the EU AI Act accessibility provisions, ADA Title II digital accessibility final rule (2024), European Accessibility Act (2025), and analogous standards in the UK, Australia, Singapore, and China. Critically, the EU AI Act classifies AI systems used in employment, credit, and public services as "high-risk," imposing mandatory WCAG-equivalent accessibility requirements as a compliance condition—not a best practice. Pivotal Accessibility [2025] establish that traditional WCAG audits are insufficient for AI-generated content because AI outputs are dynamic, not static: a page that passes WCAG at audit time may fail when AI-generated elements are rendered. This dynamic failure mode is a primary driver of the 2025–2026 WCAG failure rate reversal documented by WebAIM [2026].

Gap this paper fills. To our knowledge, no prior published work has (1) formally defined stranded capital as a demand-side analog to observed AI labor market exposure; (2) constructed a sector-level index measuring the intersection of platform inaccessibility, PWD spending power, and digital commerce penetration; or (3) empirically tested whether AI-driven page complexity growth is producing a net temporal acceleration of disability market exclusion. This paper fills all three gaps.

3 Methodology

3.1 Research Questions and Null Hypothesis

RQ1 (Construct Validity): Do AI-mediated platform sectors with higher WCAG non-compliance rates show higher SCI scores that predict observed PWD transaction dropout rates, controlling for income, age, and disability type?

RQ2 (First-Mover Premium): Do firms scoring in the top quartile of the Disability Equality Index show revenue and net income premiums consistent with a first-mover accessibility advantage, and does this premium increase in AI-intensive sectors as measured by the Massenkoff and McCrory [2026] occupational exposure scores?

RQ3 (Temporal Acceleration): Has AI-driven page complexity growth between 2019 and 2026 produced a net increase in SCI scores across the WebAIM longitudinal series, constituting a measurable temporal stranding acceleration?

Null Hypothesis (H_0): WCAG compliance level has no significant relationship with PWD transaction completion rates after controlling for income and disability type. The SCI has no predictive validity against observed click-away or transaction dropout proxies. There is no systematic temporal increase in SCI scores across the 2019–2026 WebAIM longitudinal series.

3.2 Stranded Capital Index: Formula and Derivation

The Stranded Capital Index for sector s is defined as:

$$\mathrm{SCI}_s = W_s \times D_s \times P_s \tag{1}$$

where:

- W_s = WCAG non-compliance rate for sector s (proportion of homepages with ≥ 1 detectable WCAG 2 failure, from WebAIM Million 2026)
- D_s = estimated PWD disposable income share for sector s (proportion of sector's addressable market attributable to disability population, derived from American Institutes for Research 2018 and Bureau of Labor Statistics 2025)
- P_s = digital commerce penetration for sector s (proportion of sector transactions conducted via digital platforms, from sector-level e-commerce data)

Higher SCI values indicate greater suppression of disability market spending by platform inaccessibility. The index is designed to be replicable, annually updatable as WebAIM and BLS data refresh, and extensible to firm-level analysis as accessibility audit data becomes available at that granularity.

3.3 Data Sources

We construct the SCI dataset from four primary public sources:

1. **WebAIM Million 2019–2026** [WebAIM, 2026, 2025]: Annual WCAG compliance audit of the top 1,000,000 website homepages, stratified by industry sector. Provides W_s and the longitudinal series for C3. The 2026 report (February 2026) is the primary cross-sectional dataset.

2. **BLS Current Population Survey Disability Supplements** [Bureau of Labor Statistics, 2025]: Annual labor force characteristics data by disability status including occupation and industry breakdowns (Tables 1–4). Provides disability population proportions by sector for D_s estimation.

3. **American Institutes for Research Hidden Market Report** [American Institutes for Research, 2018]: Disability-stratified disposable and discretionary income estimates from ACS microdata. Provides the baseline $490B US PWD disposable income figure and type-specific breakdowns used to calibrate D_s.

4. **Accenture Disability Equality Index Financial Performance Dataset** [Accenture et al., 2023]: Financial performance comparison across 346 US companies (2015–2022) stratified by DEI leader vs. non-leader status. Used for RQ2 first-mover premium analysis.

Supplementary sources include Return on Disability Group [2024] for global market sizing validation, TriplePundit / Safman [2025] for US click-away proxy estimates, and Clifford Chance [2024] for regulatory adjustment parameters in C3.

3.4 Analytical Conditions

We compute six variants of the SCI, each representing a distinct analytical lens on the stranded capital phenomenon:

Novel condition: Temporal Stranding Acceleration (C6). C6 is Elizabeth Hannan's original contribution and has no precedent in published AI economics or accessibility research. The theoretical basis derives from two documented dynamics in the WebAIM longitudinal data: (1) page complexity has grown 61% over six years as AI-generated content and JavaScript frameworks proliferate [WebAIM, 2026]; and (2) WCAG compliance improvement has been marginal across the same period, with the 2025–2026 year posting a

Table 1: SCI analytical conditions: design rationale and computation specification

Cond.	Label	Specification
C1	Baseline SCI	$SCI_s = W_s \times D_s \times P_s$ (unweighted). Baseline index value for all 8 sectors.
C2	AI-Intensity Weighted	SCI multiplied by sector-level AI occupational exposure score from Massenkoff and McCrory [2026]. Tests whether AI-heavy sectors show amplified stranding.
C3	Regulatory-Adjusted	SCI discounted by sector-specific regulatory compliance pressure coefficient (ADA Title II, EAA 2025, EU AI Act). Tests whether regulatory pressure measurably reduces stranding.
C4	DEI Premium Offset	SCI adjusted by the Accenture et al. [2023] 1.6× revenue premium in DEI-leading sectors. Tests first-mover accessibility advantage as a partial stranding offset.
C5	Click-Away Validation	SCI scaled by observed PWD transaction abandonment rates (69% baseline from UK Purple Pound data). Construct validity test for RQ1.
C6	**Temporal Stranding Acceleration (Novel)**	Year-over-year change in SCI computed from WebAIM 2019–2026 longitudinal series. Tests whether AI-driven page complexity growth produces a net temporal acceleration in stranded capital.

reversal (94.8% → 95.9% failure rate) [WebAIM, 2026, 2025]. C6 formalizes the interaction between these two trends as a temporal acceleration of stranded capital: a dynamic in which AI deployment is increasing the economic cost of inaccessibility faster than regulatory and market pressure can reduce it. If C6 produces a statistically significant positive year-over-year SCI trend, it establishes that AI platforms are not a passive backdrop to accessibility progress but an active driver of accessibility regression—with direct implications for AI deployment governance standards.

3.5 Sector Sample

We compute SCI across eight industry sectors:

1. **E-Commerce** ($W_s \approx 0.97$): Shopify-powered sites average 69.6 WCAG errors per homepage; WooCommerce 75.6—both substantially above the 56.1 overall mean [WebAIM, 2026]. High digital penetration and high PWD household spending on goods makes this the highest-priority stranding sector.

2. **Financial Services** ($W_s \approx 0.95$): High AI occupational exposure score [Massenkoff and

McCrory, 2026]. 76% of disabled employees do not disclose disability status to employers [Accenture et al., 2023], suggesting underrepresentation in design testing pipelines.

3. **Healthcare** ($W_s \approx 0.96$): Highest consequence per inaccessible transaction. Patient portal and telehealth AI deployments are high-risk systems under the EU AI Act [Clifford Chance, 2024].

4. **Government** ($W_s \approx 0.89$): Lowest WCAG failure rate in sector sample, consistent with ADA Title II enforcement. Expected to show largest C1–C3 gap, validating the regulatory adjustment parameter.

5. **Education** ($W_s \approx 0.94$): AI-powered learning platforms serve a population in which only 46% of disabled students enroll in postsecondary education versus 75% of non-disabled peers [Tandfonline, 2025]. Accessibility failures compound structural educational exclusion.

6. **Entertainment** ($W_s \approx 0.97$): High digital penetration. Streaming and social media platforms are not classified as high-risk under EU AI Act, creating a regulatory gap for this high-stranding sector.

7. **Travel** ($W_s \approx 0.96$): AI-powered booking and navigation platforms. Mobility disability population has high latent demand for accessible travel infrastructure [American Institutes for Research, 2018].

8. **Technology/SaaS** ($W_s \approx 0.91$): Highest AI occupational exposure score; lowest WCAG failure rate after government. DEI participation rate highest in this sector—tests the first-mover premium hypothesis most directly [Accenture et al., 2023].

3.6 Failure Rubric

We pre-register four outcome metrics, committed to GitHub prior to any analysis:

Inter-rater reliability. All sector × condition SCI scores will be independently coded by two raters using the pre-committed rubric. Cohen's κ will be computed on a 20% random sample. Target: $\kappa \geq 0.70$. Disagreements will be adjudicated prior to statistical analysis.

3.7 Statistical Analysis Plan

Primary test (RQ1): Pearson correlation between C1 (Baseline SCI) and observed PWD click-away rates by sector ($\alpha = 0.05$, two-tailed). Effect size: Pearson r. Threshold for construct validity: $r \geq 0.60$.

Regression analysis (RQ2): OLS regression of three-year sector revenue growth on SCI baseline score, DEI participation rate, and AI intensity score. Coefficient direction on SCI baseline (expected: negative) tests the stranded capital suppression hypothesis.

Temporal analysis (RQ3): Year-over-year F4 scoring across the 2019–2026 WebAIM series. Mann-Kendall trend test for monotonic increase in SCI across years. Test for structural break at 2025–2026 (reversal year).

Cross-condition comparison: Pairwise Wilcoxon signed-rank tests between C1 and C3 (regulatory effect) and between C1 and C2 (AI-intensity amplification) across sectors. Bonferroni correction for multiple comparisons.

Survey validation: Chi-square test for independence between high-SCI vs. low-SCI sector classification and binary transaction abandonment in Hannan survey instrument ($n \geq 60$). Point-biserial correlation between SCI score and self-reported platform friction rating (1–5 scale).

Table 2: Failure rubric: metric definitions and scoring scales

Code	Scale	Metric Name	Operational Definition
F1	0/1/2	Stranding Signal	0 = no stranding evidence; 1 = partial stranding (accessible alternative suboptimal); 2 = full stranding (transaction cannot complete).
F2	0/1/2	Revenue Gap Signal	0 = no measurable revenue gap between DEI leaders/non-leaders; 1 = gap present, < 1 SD; 2 = gap present, ≥ 1 SD.
F3	0/1	Regulatory Inefficacy	0 = compliance standard reduces measured SCI; 1 = compliance standard has no measurable SCI effect.
F4	0/1/2	Temporal Acceleration	0 = no SCI increase over time; 1 = SCI increasing $< 10\%$ p.a.; 2 = SCI increasing $\geq 10\%$ p.a. (C6 primary metric).

4 Preliminary Results

Work in progress. Full results will be reported upon completion of the sector-level SCI computation, inter-rater reliability coding, and survey instrument administration. This section documents the anticipated reporting structure and presents the C6 temporal trend data currently available from the WebAIM longitudinal record.

4.1 Anticipated Reporting Structure

Results will be reported in five subsections:

1. **Baseline SCI by sector (C1).** A table of SCI_s values for all eight sectors, with component values (W_s, D_s, P_s) reported separately. Expected ordering: e-commerce > entertainment > travel > healthcare > financial services > education > technology/SaaS > government.

2. **SCI under AI-weighting and regulatory adjustment (C2, C3).** Paired comparison of C1 vs. C2 (AI amplification) and C1 vs. C3 (regulatory reduction). Expected finding: C2 increases SCI most for financial services and technology/SaaS (highest AI intensity). C3 reduces SCI most for government (strongest regulatory pressure).

3. **First-mover premium analysis (C4, RQ2).** OLS regression table: SCI baseline → sector revenue growth. Expected finding: negative coefficient on SCI baseline, positive coefficient on DEI participation rate, consistent with Accenture et al. [2023].

4. **Construct validity: click-away validation (C5, RQ1).** Pearson r between C1 SCI and sector-estimated click-away rates. Expected: $r \geq 0.60$. Survey chi-square results for transaction abandonment by SCI sector group.

5. **Temporal stranding acceleration (C6, RQ3).** F4 scores across 2019–2026 WebAIM series. The 2025–2026 reversal (WCAG failure rate increasing from 94.8% to 95.9%, average errors per page from 51.0 to 56.1, representing a 10.1% year-over-year increase) provides initial support for the C6 hypothesis: AI-driven complexity growth has resumed after one year of improvement, and the acceleration rate (+10.1%) meets the F4 = 2 threshold. Mann-Kendall trend test results will be reported.

A pre-registration of hypotheses and analysis plan is archived on the Open Science Framework (see Section 7).

5 Discussion

Stranded capital as a first-class economic variable. The central argument of this paper is not that AI accessibility failures are ethically regrettable but that they are *economically consequential in a measurable, sector-specific, and policy-tractable way.* The SCI operationalizes this claim by bridging three data streams that have never been formally combined: WCAG compliance audit data, disability population income estimates, and digital commerce penetration. What emerges is not an equity metric but an economic suppression metric—one that can be trended over time, decomposed by sector, and validated against observed consumer behavior. The observed vs. theoretical exposure framework of Massenkoff and McCrory [2026] provides the intellectual license for this move: if AI economists accept that the gap between theoretical and observed labor augmentation is a real and measurable economic variable, they must accept the same for the gap between theoretical and observed disability market participation. The SCI is that variable for the demand side.

The temporal dimension: AI as an accessibility regression driver. Prior accessibility research has treated the slow pace of WCAG compliance improvement as a failure of will or enforcement—a problem of insufficient regulatory pressure. The C6 analysis suggests a structural alternative: AI deployment is an *active driver* of accessibility regression, not merely a passive backdrop to it. As AI-generated content, dynamic JavaScript frameworks, and AI-powered UI components proliferate, page complexity grows faster than compliance remediation can follow. The 2026 WebAIM reversal—the first WCAG failure rate increase in six years, occurring in the same year that EU AI Act provisions began applying and ADA Title II digital rules took effect—is precisely the dynamic the C6 hypothesis predicts. If the Mann-Kendall trend test confirms a statistically significant positive trend in SCI over the 2019–2026 period, it will constitute the first empirical evidence that AI deployment is not merely failing to serve the disability market but is actively enlarging the economic exclusion gap.

First-mover premium and the business case for SCI disclosure. Accenture et al. [2023] document the financial performance premium for disability-inclusive firms, but do not identify a mechanism. The SCI offers a partial mechanism: firms that achieve accessibility compliance face a smaller stranded capital problem—they capture transactions that competitors forfeit through inaccessibility. In AI-intensive sectors (C2), where the multiplier effect is strongest, the first-mover premium should be largest. If RQ2 confirms a negative coefficient on SCI baseline in the revenue growth regression, it will provide the first mechanism-level evidence linking accessibility compliance to financial outperformance—a finding with

direct implications for ESG materiality disclosure under SEC Item 303, GRI 203, and CSRD ESRS S1.

Policy implications and the limits of voluntary compliance. The C3 (regulatory-adjusted) condition tests a specific policy hypothesis: whether external compliance pressure measurably reduces stranded capital. Government's expected position as the lowest-SCI sector under C3 would be consistent with ADA Title II enforcement effects—but it would also reveal the ceiling of regulatory pressure: government achieves 89% WCAG failure rate under legal mandate, which means 11% of government pages still exclude PWD users despite enforcement. If voluntary DEI participation (C4) produces larger SCI reductions than regulatory adjustment (C3) in the private sector, it would suggest that the business case for accessibility is a more effective driver than compliance mandates at current enforcement levels. This finding would argue for SEC-level materiality disclosure requirements—making the SCI a reportable financial risk factor rather than a best-practice recommendation.

6 Limitations

1. **Sector-level aggregation.** SCI is currently computed at the sector level using estimated aggregate parameters. Firm-level SCI computation requires firm-specific WCAG audit data and accessibility spending disclosures that are not currently required or standardized. Sector-level findings cannot be attributed to individual firms.

2. **PWD income share estimation.** The D_s parameter is estimated from aggregate disability income data (American Institutes for Research 2018; Bureau of Labor Statistics 2025) cross-referenced with sector-level market size estimates. Direct disability-stratified transaction data by sector does not exist in public form. Future work should partner with accessibility platform vendors to access first-party transaction completion data.

3. **Click-away proxy validity.** C5 uses sector-estimated click-away rates derived from UK Purple Pound survey data [TriplePundit / Safman, 2025] and US analogues. These are not sector-specific empirical measurements; they are extrapolations. The survey validation in Step 17 partially addresses this limitation but is subject to the sample size constraints noted below.

4. **Survey sample size.** The Hannan survey instrument validation uses an initial convenience sample (n target ≥ 60). This is adequate for directional construct validity evidence but not for definitive inference. A pre-registered, nationally representative survey is recommended as follow-up.

5. **WCAG automated detection ceiling.** WebAIM Million data is generated by automated WAVE scanning, which detects approximately 30–40% of real accessibility barriers. Manual accessibility audits would likely produce substantially higher W_s values and correspondingly higher SCI scores. Published figures are therefore conservative estimates of true stranding.

6. **Novel condition (C6) single-variable trend.** The C6 temporal acceleration hypothesis is tested against a single outcome variable (WebAIM error count per page). Confounders including changes in the composition of the top 1,000,000 homepage list, shifts in CMS market share, and macroeconomic effects on website investment are not controlled. Causal claims about AI-driven complexity growth require instrumentation beyond what the current data support.

7. Researcher positionality. The author is a professional with deep ESG and accessibility domain expertise. This expertise is a primary research asset for this question ? but readers should apply standard scrutiny to the di- rection of hypotheses and interpret preliminary results accordingly.

7 Conclusion

We present *Stranded Capital*, the first formal economic framework for measuring the suppression of disability market spending power attributable to AI platform inaccessibility. The central argument is simple and empirically tractable: the disability economy commands between $13 trillion and $23 trillion in global annual spending power, yet 95.9% of the AI-mediated platforms through which that spending must flow contain detectable accessibility barriers—barriers that are growing, not shrinking, as AI-driven page complexity accelerates.

The Stranded Capital Index (SCI) operationalizes this gap as a measurable, sector-specific, annually updatable index derived entirely from public data. The C6 Temporal Stranding Acceleration condition—our novel contribution—is the first empirical test of whether AI deployment is an active driver of accessibility regression rather than a neutral backdrop. The 2025–2026 WebAIM reversal, occurring in the year AI-generated content proliferation accelerated and EU AI Act provisions began applying, provides an alarming initial data point.

The deeper structural argument extends beyond any single sector or measurement cycle. If the Massenkoff-McCrory observed exposure framework is correct that the distribution of AI's economic benefits will be shaped by deployment choices rather than capability alone, then the SCI documents what those choices are currently producing on the demand side: a systematic transfer of market power away from 1.6 billion people whose spending capacity exists but cannot reach the platforms designed to receive it. Treating this as a design aspiration is insufficient. Measuring it, disclosing it, and holding AI deployers accountable for it is a macroeconomic requirement.

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Survey and pilot data collected under IRB Exempt status: 45 CFR 46.104(d), Category 2, February 2026.

Open Science Statement

All SCI datasets, computation scripts, scoring rubric (`scoring_rubric.json`), and sector-level analysis outputs will be released at: https://github.com/[YOUR-GITHUB-USERNAME]/Economics_Pub_5_Stranded_Capital upon submission of the complete paper. Pre-registration of hypotheses and failure rubric is archived at: https://osf.io/[OSF-PROJECT-ID-TBD]. GitHub commit timestamp for rubric pre-registration: [GITHUB-COMMIT-HASH].

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